TASK-4: Order Management using Joins

AIM: Build an Order Management system to manage customers, products, and orders, and display insights like highest order and most active customer using SQL joins and subqueries.

Algorithm:

1. Create database and tables (customers, products, orders)
2. Insert sample data into tables
3. Use JOIN query to fetch customer order history
4. Use subquery to find highest value order
5. Use subquery to find most active customer
6. Build backend APIs using Node.js
7. Fetch data using JavaScript (fetch API)
8. Display results in HTML table

Code:

Index.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Order Management</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
<h1>Order History</h1>
<button onclick="loadOrders()">Load Orders</button>
<table id="ordersTable"></table>
<h2>Insights</h2>
<button onclick="highestOrder()">Highest Order</button>
<p id="highest"></p>

<button onclick="activeCustomer()">Most Active Customer</button>
<p id="active"></p>
```

```
<script src="script.js"></script>
</body>
</html>
```

Style.css

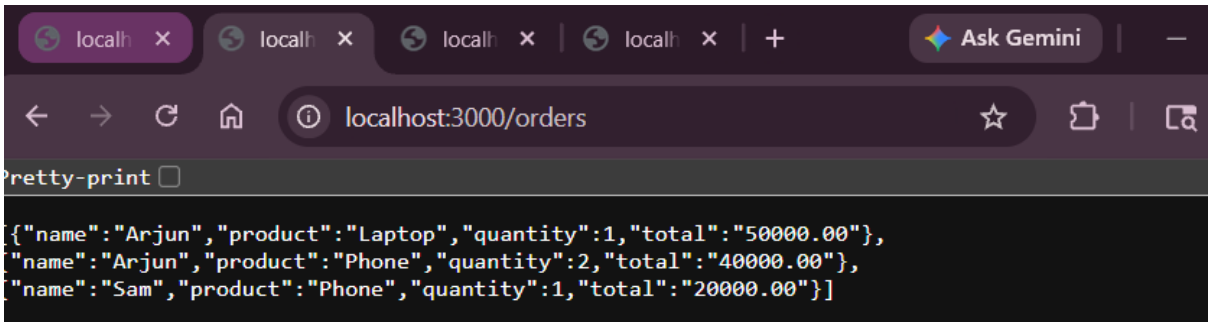
```
body {
    font-family: Arial;
    text-align: center;
}
table {
    margin: auto;
    border-collapse: collapse;
}
td, th {
    border: 1px solid black;
    padding: 10px;
}
button {
    margin: 10px;
    padding: 10px;
}
```

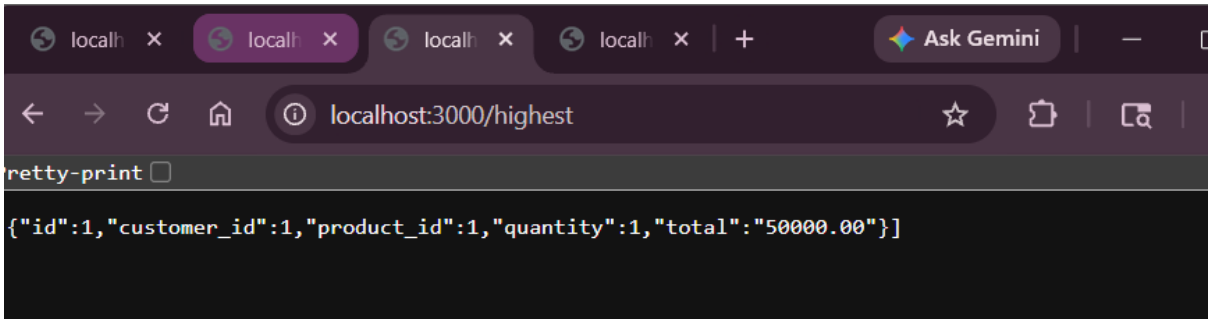
Script.js

```
function loadOrders() {
    fetch("http://localhost:3000/orders")
    .then(res => res.json())
    .then(data => {
        let table = "<tr><th>Name</th><th>Product</th><th>Qty</th><th>Total</th></tr>";
        data.forEach(o => {
            table += `<tr>
                <td>${o.name}</td>
                <td>${o.product}</td>
                <td>${o.quantity}</td>
```

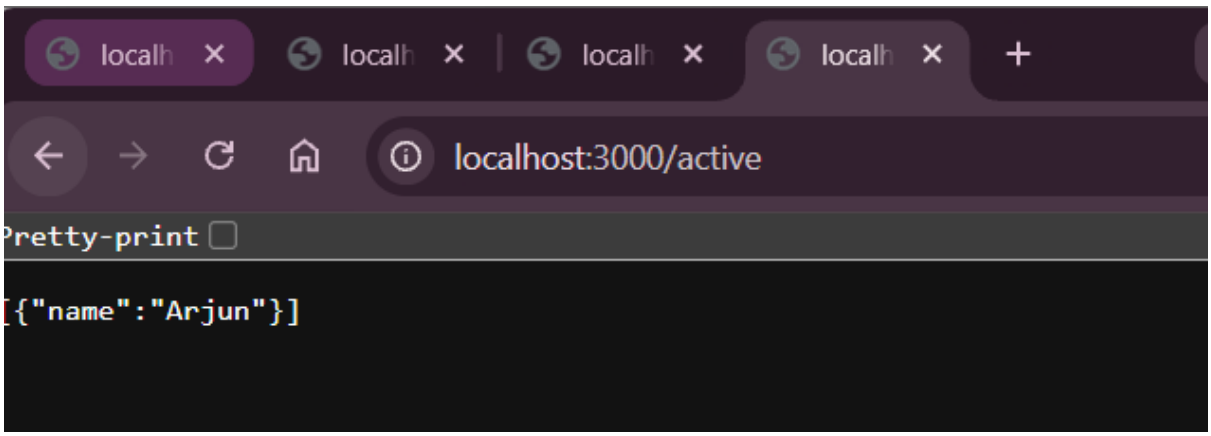
```
            <td>${o.total}</td>
          </tr>`;
        });
        document.getElementById("ordersTable").innerHTML = table;
    });
}
function highestOrder() {
    fetch("http://localhost:3000/highest")
    .then(res => res.json())
    .then(data => {
        document.getElementById("highest").innerText =
            "Highest Order: ₹" + data[0].total;
    });
}
```

OUTPUT1: Highest value order





OUTPUT2: Most active customer



Result: System displays order details using JOIN queries.
It also shows highest order and most active customer using subqueries.